Raazia Assigned Source Review

Operating Systems CEP Topic 23 | Studies S008 to S014 | 21 October 2026

This deliverable records an AI-assisted source review completed under Raazia Imran Reshamwala's direction. It contains seven source identity checks, 42 eligibility-criterion assessments, 70 QA appraisals, 13 outcome checks, a correction register and reviewed bibliography. The integrated report and master workbook include the supported corrections.

All 13 retained numerical values were found in the accessed sources. Corrections concern page anchors, software details, runtime-mode distinctions, offered-load interpretation and claim-specific QA rationale. Original scores and records remain in the frozen baseline.

Study	Original QA	Reviewed QA	Outcome rows
S008 GSLICE	17	17	2
S009 Serving DNNs like Clockwork	16	18	2
S010 Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS)	17	17	1
S011 ParvaGPU	14	14	2
S012 Torpor	15	15	2
S013 StreamBox	15	17	2
S014 Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON)	16	16	2

QA totals change from 110/140 to 114/140. Clockwork and StreamBox each gain two points because reported configuration and clearly delimited performance/trust evidence satisfy Q4 and Q9. A score is a source appraisal, not a performance measure or a human independent vote.

Version boundary: ParvaGPU values were checked against arXiv2409.14447v1. Conference publication identity is confirmed; equivalence of the numerical manuscript to the final published full text is not established. Global review scope is unchanged: 559 retained records have not received source assessment. These are study-selection limitations, not findings that this seven-study check can erase.

Integrated corrections

R01 P008 Reading copy

Original: Saved local PDF unreadable

Reviewed treatment: Recovered a readable 15-page academic-mirror copy; kept old bytes separately

Evidence: File parsing and title/DOI checks

R02 P009 E033/E034 Evidence anchor

Original: Author-copy page10 inherited

Reviewed treatment: Use accessed USENIX PDF pp10-11, printed451-452

Evidence: Figure5 and section6.1

R03 P009 Runtime extraction

Original: Software versions not assessed

Reviewed treatment: TVMv0.7 reported; preserve missing OS/driver distinctions

Evidence: PDF pp4,7; Table1; section5

R04 P009 Q3 QA rationale

Original: Repetition count unclear

Reviewed treatment: Scaling experiment has three repetitions; retained local Figure5 repetitions unspecified

Evidence: PDF p14 section6.6; pp10-11 section6.1

R05 P009 Q4/Q9 QA reappraisal

Original: 1/2 for both

Reviewed treatment: 2/2 for both: hardware/compiler/workload documented; performance isolation evaluated within explicit boundary

Evidence: Submitted rubric Table8; source anchors in QA review

R06 P013 Q4/Q9 QA reappraisal

Original: 1/2 for both

Reviewed treatment: 2/2 for both: hardware/framework/workload documented; trust assumptions and scope explicitly delimited

Evidence: Submitted rubric Table8; source anchors in QA review

R07 P012 Isolation/fault extraction

Original: Not assessed in retained extract

Reviewed treatment: Section4.5 describes two runtime modes and recovery; section7 uses sharing mode, section8 isolation mode; direct fault-test evidence not established

Evidence: PDF pp7,9; sections4.5,7-8

R08 P014 E042 Offered-load context

Original: Poisson rate half maximum in shared test

Reviewed treatment: Initial rate half solo maximum; shared test halves it again for50% allocation

Evidence: PDF pp12-13 section5.5

R09 P009/P012/P014 Q10 QA rationale

Original: Artifact/reproduction wording used inconsistently

Reviewed treatment: Appraise source replication information; do not penalize a paper merely because this review did not rerun it

Evidence: Protocol Q10; revised QA rationales

R10 P008-P014 Bibliography

Original: Generic misc entries and abbreviated authors

Reviewed treatment: Create full-author conference entries; label ParvaGPU accessed manuscript separately

Evidence: Accessed PDF title pages and SC24/author record

R11 P011 Version record

Original: Published/preprint relationship uncertain

Reviewed treatment: Publication identity confirmed; numerical check remains explicitly scoped to arXivv1

Evidence: SC24 official record and accessed arXivv1

R12 Review attribution Reviewer identity

Original: User requested assistant to do assigned work

Reviewed treatment: Record AI assistant under Raazia direction; independent human vote left blank

Evidence: Actual review method

S008 GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform

Report P008 | 2020 | SoCC 2020

DOI or proceedings: 10.1145/3419111.3421284

Source: https://www.cse.iitb.ac.in/~puru/courses/spring2024-25/downloads/gslice.pdf

Version checked: Conference paper; title and author identity matched in accessed PDF

PDF pages: 15 | SHA256: f8e67a94189a3db224886768bd197e020f30a26572caf7f88cbb28efd35631c9

Eligibility appraisal

Criterion	Finding and basis
IC1	Supported by accessed conference copy. Conference title/authors/year matched in full text. Anchor: PDF title/cover page
IC2	Supported. MPS allocation among concurrent inference functions Anchor: PDF pp4-8; section3
IC3	Supported. Inference workload is explicit; training/rendering outcomes are kept separate where present. Anchor: PDF p9; section4.2; Figure10
IC4	Supported. Retained outcome values and methods are inspectable in the accessed source. Anchor: GSLICE PDF p9; printed p500; Table3; GSLICE PDF p10; printed p501; Table4 and section4.3
IC5	Supported. Extracted evidence concerns co-tenancy or control on one physical GPU; fleet-only outcomes not substituted. Anchor: PDF p9; section4.2; Figure10
IC6	Supported. 2020 is within the submitted2020-2026 window. Anchor: Conference title page / source identity

Quality appraisal

Item	Original	Reviewed	Rationale and source anchor
Q1	2	2	Inference sharing objective explicit. PDF p1; printed p492; abstract/introduction
Q2	2	2	MPS allocation and adaptive batching described. PDF pp4-8; section3
Q3	1	1	Confidence intervals shown; repeated-run count not specified for retained results. PDF p9; printed p500; section4.1
Q4	2	2	V100 testbed, kernel, frameworks and concurrent models specified. PDF pp8-9; section4 setup
Q5	2	2	TensorRT comparison matches GPU-ready batch execution and states excluded transport costs. PDF p10; printed p501; section4.3
Q6	2	2	GUE defined as an efficacy metric; table values are not busy percentages. PDF p3; printed p494; PDF p9 Table3
Q7	1	1	Median latency and SLO shown; request-tail quantile not established. PDF p10; section4.4; Figures12a-12b
Q8	2	2	Two-to-four-function combinations evaluated. PDF p9; section4.2; Figure10
Q9	2	2	Performance boundary evaluated under allocated GPU shares; no

Item	Original	Reviewed	Rationale and source anchor
			adversarial guarantee inferred. PDF pp5-7,10-11; sections3.2,4.5
Q10	1	1	Implementation and sensitivity described; full reproduction package not established in accessed copy. PDF pp8-13; sections4-6

Verified reported outcomes

E031 GPU utilization efficacy GUE: 94.04 composite score.

Baseline: Default MPS GUE38.82. Workload: Alexnet+ResNet50;25ms target. Hardware: One V10016GB.

Interpretation: Not percent GPU utilization Anchor: GSLICE PDF p9; printed p500; Table3.

E032 Signed latency improvement: -7.52 percent.

Baseline: TensorRT server19.02. Workload: ResNet50;batch1;GPU-ready batch execution. Hardware: One V10016GB.

Interpretation: Negative means worse latency; excludes transport overhead; single-model comparison Anchor: GSLICE PDF p10; printed p501; Table4 and section4.3.

S009 Serving DNNs like Clockwork: Performance Predictability from the Bottom Up

Report P009 | 2020 | USENIX OSDI 2020

DOI or proceedings: USENIX proceedings (no DOI assigned)

Source: https://www.usenix.org/system/files/osdi20-gujarati.pdf

Version checked: Conference paper; title and author identity matched in accessed PDF

PDF pages: 21 | SHA256: 31e12ad02b25dfc0296bbaa577c4dd46cee8bf3ae4e4db51aaa5d557a794abfb

Eligibility appraisal

Criterion	Finding and basis
IC1	Supported by accessed conference copy. Conference title/authors/year matched in full text. Anchor: PDF title/cover page
IC2	Supported. Temporal scheduling of multiple inference models on one GPU Anchor: PDF pp6-10; sections4-5
IC3	Supported. Inference workload is explicit; training/rendering outcomes are kept separate where present. Anchor: PDF pp10-12; sections6.1-6.3; Figures5-7
IC4	Supported. Retained outcome values and methods are inspectable in the accessed source. Anchor: USENIX PDF pp10-11; printed pp451-452; Figure5 and section6.1; USENIX PDF pp10-11; printed pp451-452; Figure5 and section6.1
IC5	Supported. Extracted evidence concerns co-tenancy or control on one physical GPU; fleet-only outcomes not substituted. Anchor: PDF pp10-12; sections6.1-6.3; Figures5-7
IC6	Supported. 2020 is within the submitted2020-2026 window. Anchor: Conference title page / source identity

Quality appraisal

Item	Original	Reviewed	Rationale and source anchor
Q1	2	2	Predictable multi-model inference objective and target behavior explicit. PDF p2; printed p443; section1
Q2	2	2	Controller scheduling, worker actions and managed model runtime described. PDF pp6-10; sections4-5
Q3	1	1	Section6.6 reports three scaling repetitions; repeated-run count for retained Figure5 is not specified. PDF pp10-11,14; sections6.1,6.6
Q4	1	2	V100 hardware, TVM v0.7 compiler, model set and client configuration are reported. PDF pp4,7,10; Table1; sections2,6
Q5	1	1	INFaaS uses different EC2 machines; authors restrict that contrast to qualitative interpretation. PDF p10; printed p451; section6.1
Q6	2	2	Goodput counts successful requests within their deadline and excludes late requests. PDF p11; printed p452; section6.1
Q7	2	2	SLO range, closed-loop load, latency CDF and batch scheduling explanation are explicit. PDF pp10-11; Figure5; section6.1
Q8	2	2	Concurrent model demand and a protected minor workload tested. PDF pp10-12; sections6.1-6.3; Figures5-7

Item	Original	Reviewed	Rationale and source anchor
Q9	1	2	Performance-isolation boundary is measured; dedicated-worker assumptions and security exclusions are explicit. PDF pp11-12,14-15; sections6.2,7
Q10	2	2	Runtime, scheduling, model details and limitations support informed implementation appraisal; no local reproduction claimed. PDF pp7-10,14-15; sections5,7

Verified reported outcomes

E033 Clockwork goodput: approximately800 requests/second.

Baseline: INFaaS on nonidentical EC2 setup. Workload: 15 ResNet50 instances;16 clients/model;500ms SLO. Hardware: One V10032GB in section6.1.

Interpretation: Within-SLO successes; comparator qualitative; not total throughput Anchor: USENIX PDF pp10-11; printed pp451-452; Figure5 and section6.1.

E034 Clockwork request latency: approximately300 milliseconds.

Baseline: No comparative ratio. Workload: 15 ResNet50 instances;16 clients/model;500ms SLO. Hardware: One V10032GB in section6.1.

Interpretation: Batch16 scheduling explanation; not a p99 value Anchor: USENIX PDF pp10-11; printed pp451-452; Figure5 and section6.1.

S010 Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS)

Report P010 | 2025 | USENIX ATC2025, pp1657-1675

DOI or proceedings: USENIX proceedings (no DOI assigned)

Source: https://www.usenix.org/system/files/atc25-wang-jiali.pdf

Version checked: Conference paper; title and author identity matched in accessed PDF

PDF pages: 20 | SHA256: a479c6a6719d7577e36c31c1ee216a6ca8f6cadb5b4b09b1ff73ee6f6befaca0

Eligibility appraisal

Criterion	Finding and basis
IC1	Supported by accessed conference copy. Conference title/authors/year matched in full text. Anchor: PDF title/cover page
IC2	Supported. Single-GPU inference/training co-location with separable inference P99 Anchor: PDF pp5-10; sections3-5
IC3	Supported. Inference workload is explicit; training/rendering outcomes are kept separate where present. Anchor: PDF p11; Figure10
IC4	Supported. Retained outcome values and methods are inspectable in the accessed source. Anchor: USENIX PDF p11; printed p1666; Figure10 discussion
IC5	Supported. Extracted evidence concerns co-tenancy or control on one physical GPU; fleet-only outcomes not substituted. Anchor: PDF p11; Figure10
IC6	Supported. 2025 is within the submitted2020-2026 window. Anchor: Conference title page / source identity

Quality appraisal

Item	Original	Reviewed	Rationale and source anchor
Q1	2	2	Inference-priority co-location objective and training progress goal explicit. PDF p2; printed p1657; abstract/introduction
Q2	2	2	VMM pool, handover, memory mapping and SM allocation described. PDF pp5-10; sections3-5
Q3	1	1	Trace distributions and 300-second window specified; repeated-run uncertainty not specified for Figure10. PDF pp10-11; section6; Figure10
Q4	2	2	V100/A100 cases, CPU/RAM, software versions and model/batch table supplied. PDF pp10-11,14; Table3; sections6,6.6
Q5	1	1	TaskSwitch is an adapted implementation; equivalence to unmodified PipeSwitch is not established. PDF p10; comparing targets
Q6	2	2	Training samples/second and inference SLO compliance are defined separately. PDF p11; metrics; Figure10
Q7	2	2	P99, arrivals, duration and four-times-solo SLO definition stated. PDF p11; section6.1
Q8	2	2	Inference plus Swin-T training co-location evaluated on a single GPU. PDF p11; Figure10

Item	Original	Reviewed	Rationale and source anchor
Q9	1	1	Handover correctness and priority boundary described; direct fault-injection validation not established. PDF pp6-10; sections4-5
Q10	2	2	Artifact appendix reports availability, functionality and reproduction badges with artifact DOI. PDF p20; artifact appendix; DOI10.5281/zenodo.15581800

Verified reported outcomes

E035 Average P99 inference latency improvement factor, as reported: 12.6 x.

Baseline: TaskSwitch (authors' adapted temporal-switch baseline). Workload: section6.1 single-V100 DNN inference+Swin-T; Light/Heavy/Burst/Skewed/MAF traces;300s. Hardware: single NVIDIA Tesla V100 16GB in section6.1; four-V100 host; separate section6.6 single A100 80GB LLM experiment.

Interpretation: Author-reported average across tested workloads; not a uniform per-model speedup, and not a comparison to unmodified PipeSwitch. Anchor: USENIX PDF p11; printed p1666; Figure10 discussion.

S011 ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments

Report P011 | 2024 | SC2024 identity; extracted text is arXiv2409.14447v1

DOI or proceedings: 10.1109/SC41406.2024.00048

Source: https://arxiv.org/pdf/2409.14447

Version checked: arXiv2409.14447v1; SC24 publication identity confirmed from conference/author records; final text equivalence not established

PDF pages: 12 | SHA256: fb48e7b3e15cdedd1e9abec9b69764a9e18531fe3346e63d92d445bdf72c8f50

Eligibility appraisal

Criterion	Finding and basis
IC1	Final-version condition open. SC24/author records confirm peer-reviewed publication, but numerical source is arXivv1 and final-text equivalence is not established. Anchor: SC24 official record; author publication page; arXivv1 title page
IC2	Supported. MPS processes within a MIG instance Anchor: arXiv v1 PDF pp4-8; sectionIII
IC3	Supported. Inference workload is explicit; training/rendering outcomes are kept separate where present. Anchor: arXiv v1 PDF pp4-5; Figures3-4
IC4	Supported. Retained outcome values and methods are inspectable in the accessed source. Anchor: arXiv v1 PDF p5; sectionIII-B; Figures3-4 on p4; arXiv v1 PDF p5; sectionIII-B; Figures3-4 on p4
IC5	Supported. Extracted evidence concerns co-tenancy or control on one physical GPU; fleet-only outcomes not substituted. Anchor: arXiv v1 PDF pp4-5; Figures3-4
IC6	Supported. 2024 is within the submitted2020-2026 window. Anchor: Conference title page / source identity

Quality appraisal

Item	Original	Reviewed	Rationale and source anchor
Q1	2	2	SLO-constrained inference resource objective explicit. arXiv v1 PDF pp1-2
Q2	2	2	MIG/MPS profiling, segment configuration and allocation algorithms described. arXiv v1 PDF pp4-8; sectionIII
Q3	1	1	Profiler and six scenarios specified; repeated-run uncertainty not established. arXiv v1 PDF pp5,8-10; sectionsIII-C,IV
Q4	1	1	A100, Ubuntu and CUDA reported; printed PyTorch1.14.0 string is an unresolved environment ambiguity. arXiv v1 PDF p8; sectionIV-A
Q5	1	1	Local process-count contrast interpretable; fleet baseline variants and manuscript scope constrain comparison. arXiv v1 PDF pp5,8; sectionsIII-B,IV-A
Q6	2	2	Local throughput values and metric units given. arXiv v1 PDF pp4-5; Figure3; sectionIII-B
Q7	1	1	Local latency values given without a named tail quantile. arXiv v1 PDF

Item	Original	Reviewed	Rationale and source anchor
			pp4-5; Figure4; sectionIII-B
Q8	2	2	One-to-three MPS processes measured inside a MIG segment. arXiv v1 PDF pp4-5; Figures3-4
Q9	1	1	MIG/MPS boundary described; no dedicated fault or adversarial experiment established in reviewed manuscript. arXiv v1 PDF pp2-4,8-10
Q10	1	1	Algorithms and reconfiguration limitation described; accessed manuscript has no verified final-version equivalence. arXiv v1 PDF pp5-10; sectionIII-F; source/version record

Verified reported outcomes

E036 Reported throughput: 446 requests/s.

Baseline: same configuration with1 process (354requests/s). Workload: InceptionV3,1-GPC MIG instance,batch4;3 MPS processes. Hardware: A10080GB; Amazon p4de.24xlarge hosts with8 GPUs; extracted profiler point concerns one MIG segment.

Interpretation: One local configuration, not fleet savings. Source is arXivv1; final published version not verified. Anchor: arXiv v1 PDF p5; sectionIII-B; Figures3-4 on p4.

E037 Reported inference latency; quantile unspecified: 27 ms.

Baseline: same configuration with1 process (11ms). Workload: InceptionV3,1-GPC MIG instance,batch4;3 MPS processes. Hardware: A10080GB; Amazon p4de.24xlarge hosts with8 GPUs; extracted profiler point concerns one MIG segment.

Interpretation: Quantile/arrival model not established for this profiler point; do not label p99. Value is specific to the accessed author manuscript. Anchor: arXiv v1 PDF p5; sectionIII-B; Figures3-4 on p4.

S012 Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference

Report P012 | 2025 | USENIX ATC2025 pp597-612

DOI or proceedings: USENIX proceedings (no DOI assigned)

Source: https://www.usenix.org/system/files/atc25-yu.pdf

Version checked: Conference paper; title and author identity matched in accessed PDF

PDF pages: 17 | SHA256: f23e994fea5b089e01ecb7d7d03b371ca5d74a64f4c3a2aa9386a146eb2e5958

Eligibility appraisal

Criterion	Finding and basis
IC1	Supported by accessed conference copy. Conference title/authors/year matched in full text. Anchor: PDF title/cover page
IC2	Supported. Many low-rate inference functions temporally share one GPU Anchor: PDF pp5-9; sections4-6
IC3	Supported. Inference workload is explicit; training/rendering outcomes are kept separate where present. Anchor: PDF pp10-11; Figure8; section7.2
IC4	Supported. Retained outcome values and methods are inspectable in the accessed source. Anchor: USENIX PDF pp10-11; printed pp605-606; Figure8 and section7.2; USENIX PDF pp10-11; printed pp605-606; default P98 metrics and section7.2
IC5	Supported. Extracted evidence concerns co-tenancy or control on one physical GPU; fleet-only outcomes not substituted. Anchor: PDF pp10-11; Figure8; section7.2
IC6	Supported. 2025 is within the submitted2020-2026 window. Anchor: Conference title page / source identity

Quality appraisal

Item	Original	Reviewed	Rationale and source anchor
Q1	2	2	Low-frequency serverless inference density and latency objectives explicit. PDF p2; printed p597; section1
Q2	2	2	Remoting, swapping, memory manager, scheduler and two runtime modes described. PDF pp5-9; sections4-6
Q3	1	1	Trace, rates and warm-up specified; repeated-run uncertainty not specified for retained Figure8. PDF pp9-11; section7
Q4	1	1	Worker and model configuration supplied; exact OS/driver/framework version matrix not specified in setup. PDF p9; printed p604; settings
Q5	1	1	Native and adapted INFless-KA baselines described; CPU execution paths and adaptations affect equality. PDF pp10-11; sections7.1-7.3
Q6	2	2	Normalized single-GPU throughput and GPU busy-time definition supplied. PDF pp10-11; metrics; Figure8
Q7	2	2	Default P98, per-function rates and deadlines stated. PDF pp10-11; metrics; section7.2
Q8	2	2	Many low-frequency functions share one GPU; node-level results

Item	Original	Reviewed	Rationale and source anchor
			identified separately. PDF pp10-11; Figure8; section7.2
Q9	1	1	Runtime-sharing and runtime-isolation modes plus recovery design stated; dedicated fault-injection evaluation not established. PDF p7; section4.5; PDF p9 evaluation mode
Q10	1	1	Implementation and mode trade-offs explained; complete public replication recipe not established in paper. PDF pp7-9,13; sections4.5,6,8

Verified reported outcomes

E038 Reported single-GPU throughput factor: >10 x.

Baseline: Native early binding at same offered function rate. Workload: multiple ResNet152 functions on one V10032GB;10requests/minute/function. Hardware: single V10032GB in local Figure8; evaluation worker has4 V10032GB GPUs.

Interpretation: Lower bound reported by authors, not an exact10x; function count varies to saturate memory. No fleet cost claim. Anchor: USENIX PDF pp10-11; printed pp605-606; Figure8 and section7.2.

E039 Reported P98 latency upper bound: <50 ms.

Baseline: Native under matched request-rate setting. Workload: same single-GPU low-rate ResNet152 test,10-80requests/minute/function. Hardware: single V10032GB in local Figure8; evaluation worker has4 V10032GB GPUs.

Interpretation: Bound from text, not a digitized point; default98th percentile from section7 metrics. Applies to this workload. Anchor: USENIX PDF pp10-11; printed pp605-606; default P98 metrics and section7.2.

S013 StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow

Report P013 | 2024 | USENIX ATC2024 pp59-73

DOI or proceedings: USENIX proceedings (no DOI assigned)

Source: https://www.usenix.org/system/files/atc24-wu-hao.pdf

Version checked: Conference paper; title and author identity matched in accessed PDF

PDF pages: 16 | SHA256: d8de46489c9256a44d867b6945e909ce4fce24d7e579351c7e4247bddcfaa255

Eligibility appraisal

Criterion	Finding and basis
IC1	Supported by accessed conference copy. Conference title/authors/year matched in full text. Anchor: PDF title/cover page
IC2	Supported. Streams and runtime/API sharing across inference workflows Anchor: PDF pp5-10; sections3-8
IC3	Supported. Inference workload is explicit; training/rendering outcomes are kept separate where present. Anchor: PDF p13; section9.4; Figure19
IC4	Supported. Retained outcome values and methods are inspectable in the accessed source. Anchor: USENIX PDF p11; printed p68; section9.1; Figure13; USENIX PDF p13; printed p70; section9.3; Figure18
IC5	Supported. Extracted evidence concerns co-tenancy or control on one physical GPU; fleet-only outcomes not substituted. Anchor: PDF p13; section9.4; Figure19
IC6	Supported. 2024 is within the submitted2020-2026 window. Anchor: Conference title page / source identity

Quality appraisal

Item	Original	Reviewed	Rationale and source anchor
Q1	2	2	Workflow runtime-redundancy and spatial-sharing objective explicit. PDF p2; printed p59; section1
Q2	2	2	API proxy, PTB transformation, stream sharing and memory pool described. PDF pp5-10; sections3-8
Q3	1	1	Trace types, ablations and workloads stated; repeated-run uncertainty not specified for Figure13. PDF pp10-13; section9
Q4	1	2	V100 hardware, CPU/RAM, framework versions, CUDA version and workflow descriptions reported. PDF p10; printed p67; setup; Table2
Q5	1	1	Warm-up and common optimizations disclosed; adapted baselines and batch/concurrency choices constrain causal attribution. PDF pp10-11; comparing targets; Figure13
Q6	2	2	Throughput, memory metrics and comparator-specific factors explicitly reported. PDF p11; printed p68; Figures12-13
Q7	1	1	Traffic SLO and CDF supplied; exact scalar request-tail quantile not supplied for retained throughput result. PDF p11; Figure14
Q8	2	2	Separate three- and six-user single-GPU tests supplied. PDF p13;

Item	Original	Reviewed	Rationale and source anchor
			section9.4; Figure19
Q9	1	2	Mutual trust within workflow is explicit; separate proxies for users are distinguished from untested strong security. PDF pp5-6,13; sections3-4,9.4,10
Q10	2	2	Implementation, preparation costs and multi-user limitations sufficiently detailed for informed appraisal. PDF pp6-10,13; sections4-9

Verified reported outcomes

E040 Author-reported average throughput factor: 6.7 x.

Baseline: INFless with documented warm-up/pipelined loading/model sharing. Workload: three workflow types,PyTorch/TensorFlow,Azure-derived invocation patterns; evaluated V100 setup. Hardware: V10016GB,80CUs; host has4 GPUs; single-GPU multi-user test in section9.4.

Interpretation: Aggregate within-paper result across tested workflows; not universal or a matched cross-paper speedup. Batch/concurrency optimizations contribute. Anchor: USENIX PDF p11; printed p68; section9.1; Figure13.

E041 Reported API-forwarding overhead range: 3-5 microseconds.

Baseline: direct API path; instrumentation overhead measurement. Workload: GPU API forwarding with48concurrent functions on one GPU. Hardware: V10016GB,80CUs; host has4 GPUs; single-GPU multi-user test in section9.4.

Interpretation: Per-API overhead, not end-to-end inference latency; many APIs asynchronous. Anchor: USENIX PDF p13; printed p70; section9.3; Figure18.

S014 Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON)

Report P014 | 2025 | USENIX ATC2025 pp1003-1019

DOI or proceedings: USENIX proceedings (no DOI assigned)

Source: https://www.usenix.org/system/files/atc25-zhang-shulai.pdf

Version checked: Conference paper; title and author identity matched in accessed PDF

PDF pages: 18 | SHA256: 5b5db1111ea8aa979d383e9480bbf931ee4e9194f96f7bf17daa61ea21b64264

Eligibility appraisal

Criterion	Finding and basis
IC1	Supported by accessed conference copy. Conference title/authors/year matched in full text. Anchor: PDF title/cover page
IC2	Supported. Kernel interception and temporal/MIG sharing with separate inference measurements Anchor: PDF pp5-10; sections3-4
IC3	Supported. Inference workload is explicit; training/rendering outcomes are kept separate where present. Anchor: PDF p13; sections5.5-5.6
IC4	Supported. Retained outcome values and methods are inspectable in the accessed source. Anchor: USENIX PDF pp12-13; printed pp1013-1014; section5.5; Figure17; USENIX PDF p13; printed p1014; section5.7
IC5	Supported. Extracted evidence concerns co-tenancy or control on one physical GPU; fleet-only outcomes not substituted. Anchor: PDF p13; sections5.5-5.6
IC6	Supported. 2025 is within the submitted2020-2026 window. Anchor: Conference title page / source identity

Quality appraisal

Item	Original	Reviewed	Rationale and source anchor
Q1	2	2	Compatibility, isolation and performance-control objectives explicit. PDF p2; printed p1003; section1
Q2	2	2	Kernel interception, token control and MIG orchestration explained. PDF pp5-10; sections3-4
Q3	1	1	Setup and workloads detailed; repeated-run uncertainty not specified for retained QoS result. PDF pp10-13; section5
Q4	2	2	OS, kernel, driver, runtimes, GPUs and workloads explicitly tabulated. PDF p10; printed p1011; Tables1-2
Q5	1	1	Isolation-capable baselines and MIG adaptations described; policy and allocation differences limit equivalence. PDF pp10-11; section5.1
Q6	2	2	Throughput targets are solo-normalized and QoS violation is a request fraction. PDF pp10,12-13; sections5.1,5.4-5.6
Q7	2	2	Solo-P99 threshold, Poisson load, subsequent rate halving and token sizes stated. PDF pp12-13; section5.5; Figure17
Q8	2	2	Co-running inference relative-error and QoS experiments supplied. PDF

Item	Original	Reviewed	Rationale and source anchor
			p13; sections5.5-5.6
Q9	1	1	Baseline fault propagation is tested; design/security argument does not directly establish universal KRYPTON containment. PDF p4 Figure3; PDF p14 section7
Q10	1	1	Implementation and MIG limitation detailed; complete end-to-end replication package not established. PDF pp10,14; sections4,7

Verified reported outcomes

E042 Reported average QoS violation rate: 1.23 percent of requests.

Baseline: original soloP99 threshold (1%baseline violations by definition). Workload: Transformer-XL inference; batch8/16/32. Original rate is half solo maximum; shared test halves that rate again and assigns 50% time; tokens5-80ms.. Hardware: A10040GB for local quota/QoS tests; twoRTX4090 used for separate distributed-training test.

Interpretation: Average across tested configurations; no claim that worst-case violation is1.23%. Threshold-relative metric, not latency in milliseconds. Anchor: USENIX PDF pp12-13; printed pp1013-1014; section5.5; Figure17.

E043 Reported kernel module host-memory footprint: 4.8 MB.

Baseline: no module; additional host memory cost. Workload: kernel-space GPU sharing module on evaluated Linux host. Hardware: A10040GB for local quota/QoS tests; twoRTX4090 used for separate distributed-training test.

Interpretation: Host memory, not GPU VRAM; separate monitoring thread also required. Anchor: USENIX PDF p13; printed p1014; section5.7.

Review attribution: AI assistant under Raazia direction. Checked UTC: 2026-10-21T09:43:49.697614+00:00. Source-check records do not assert a second independent human review or experimental reproduction.